3 FIRE DETECTION

Property 3 of 7

THE PROBLEM

Most fires start small. The danger is that nobody sees them in time. Early detection gives people time to leave safely.

YOUR MISSION

Find the early signs of a fire, sound a clear alarm and help reduce the danger.

TARGETS AND LIMITS

Target	Requirement
Levels	The system shows three danger levels, for example normal, warning and alarm, on the dashboard and on the board.
Response	The alarm starts within 20 seconds after the test condition is applied.
Reliability	No alarm in normal conditions during the whole demo.
Reset	Only an allowed person can reset the alarm.

IN THE REAL WORLD

A real building has many rooms and strict safety rules. A real system is certified, tested often and must keep working when a part fails.

ADVANCED OBJECTIVES (OPTIONAL)

- Estimate the risk level and how fast it rises
- Detect a faulty sensor
- Watch two rooms on one dashboard

SAFETY RULES

No real fire, flame or smoke is allowed at the event. The organizers will show you safe ways to test.

TEAM PLANNING SHEET

Use this sheet during Build Your Strategy. You have 45 minutes. Write short and clear answers. Submit it with the first link sent to your team's email. The jury checks it at the plan check.

Each team receives two links by email. The first link is for your plan, during Build Your Strategy. The second link is for your presentation, at the end of Final Inspection, before you present to the jury.

Team name:	**Property:**
Members:	**Team leader:**
Constraint card:	

THE PROBLEM IN OUR OWN WORDS

Who is affected? What can fail? What does an incident or a loss cost?

OUR SIX LAYERS

Layer	What we will do	Who is in charge
Sensing		
Communication		
Edge processing and AI		
Actuation		
Security		
Operator interface		

OUR ARCHITECTURE

Sketch the six layers and show where each target of our property is handled (architecture diagram).

TEAM PLANNING SHEET (CONTINUED)

OUR PARTS LIST

Part	How many	Cost (credits)	Why we need it	Alternative if sold out

Total cost: _____ **Our budget:** 490 credits (440 after a re-draw) **Estimated unit cost of the real deployment:** _____

HOW OUR PROTOTYPE MAPS TO THE REAL SOLUTION

What changes, what is preserved, and why the prototype stays representative.

OUR BIGGEST RISK AND OUR PLAN B (ALSO FOR SOLD-OUT PARTS)
